

SMARTX: Porewater Sulfide

July 2026 Samples

2026-07-24

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
#And Will Write a Rerun File

###things that need to be changed
sample_month = "July"
sample_year = "2026"
Run_Date = as.Date("07/23/2026", format = "%m/%d/%Y")
Run_by = "Melanie Giessner" #Instrument user
Script_run_by = "Melanie Giessner" #Code user

Run_notes="440-80 & 450-80 not collected" #any notes from run

#Import Plate Data

folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE, pattern = ("(202607|202608)(.+Plate)"))
print(Raw_plates)

## [1] "Raw Data/20260720_SMARTX_H2S_Plate1.csv"
## [2] "Raw Data/20260720_SMARTX_H2S_Plate2.csv"
## [3] "Raw Data/20260720_SMARTX_H2S_Plate3.csv"
## [4] "Raw Data/20260720_SMARTX_H2S_Plate4.csv"
## [5] "Raw Data/20260720_SMARTX_H2S_Plate5.csv"
## [6] "Raw Data/20260722_SMARTX_H2S_Plate6.csv"
## [7] "Raw Data/20260722_SMARTX_H2S_Plate7.csv"
## [8] "Raw Data/20260722_SMARTX_H2S_Plate8.csv"
## [9] "Raw Data/20260722_SMARTX_H2S_Plate9.csv"
## [10] "Raw Data/20260724_SMARTX_H2S_Rerun1.csv"
## [11] "Raw Data/20260724_SMARTX_H2S_Rerun2.csv"

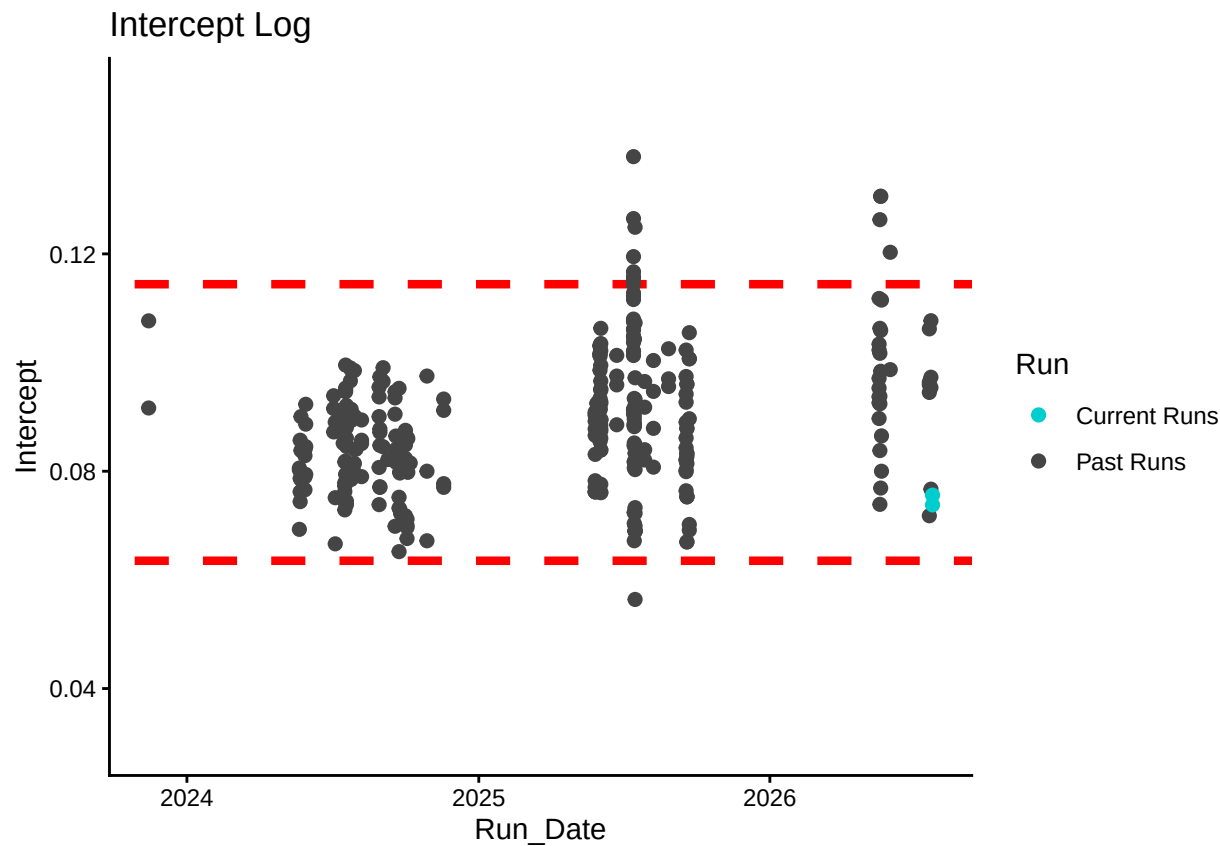
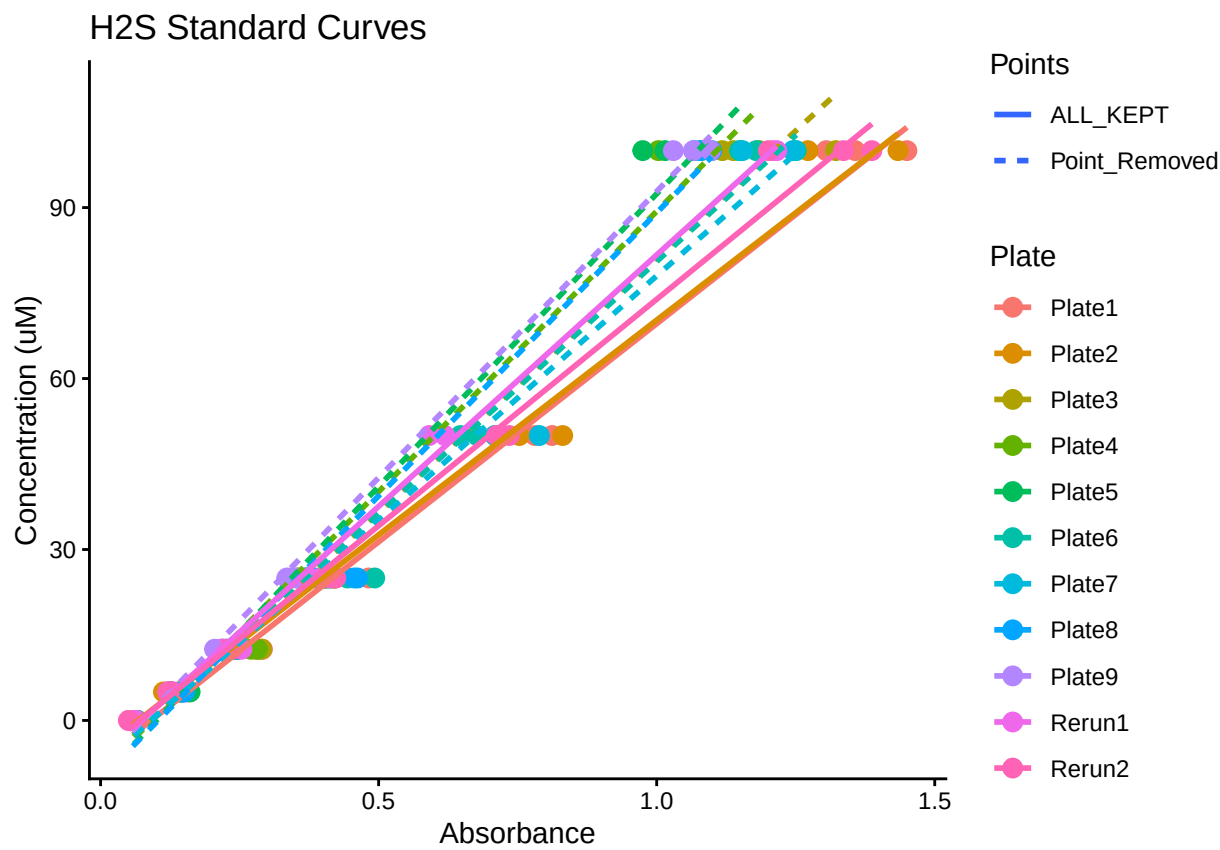
# Define the file path for QAQC log file - NO Need to change just check year
log_path <- here("Porewater QAQC Logs", "GCRew_H2S_QAQC_Log.csv")

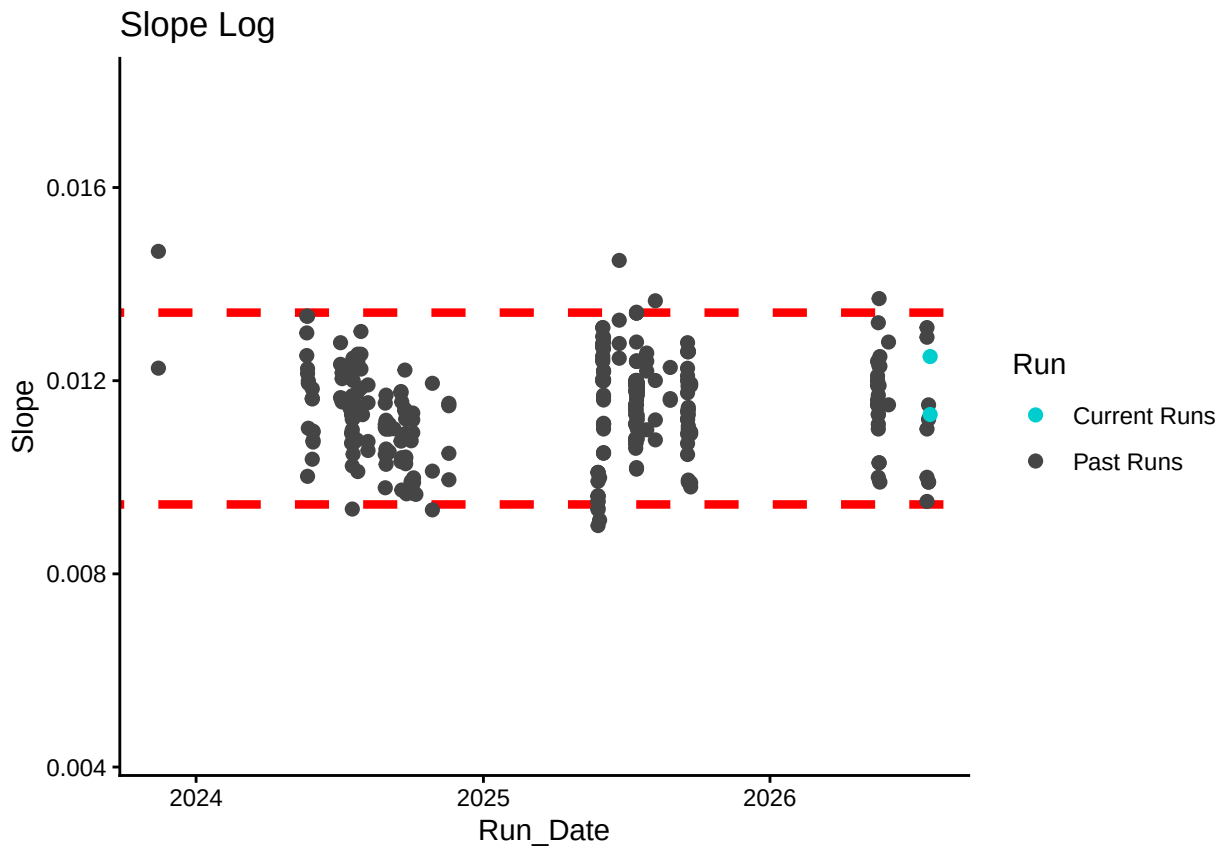
# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCRew_SMARTX_Porewater_H2S_202607.csv"
rerun_path <- "Processed Data/GCRew_SMARTX_Porewater_H2S_202607_reruns.csv"
```

```
##Read in metadata and create similar sample IDs for matching to samples
##Read in data
##Remove High CV Stds
##Check R2 & Remove Standards if Necessary

## Joining with 'by = join_by(Plate, IDs)'
```

Plot Standard Curves





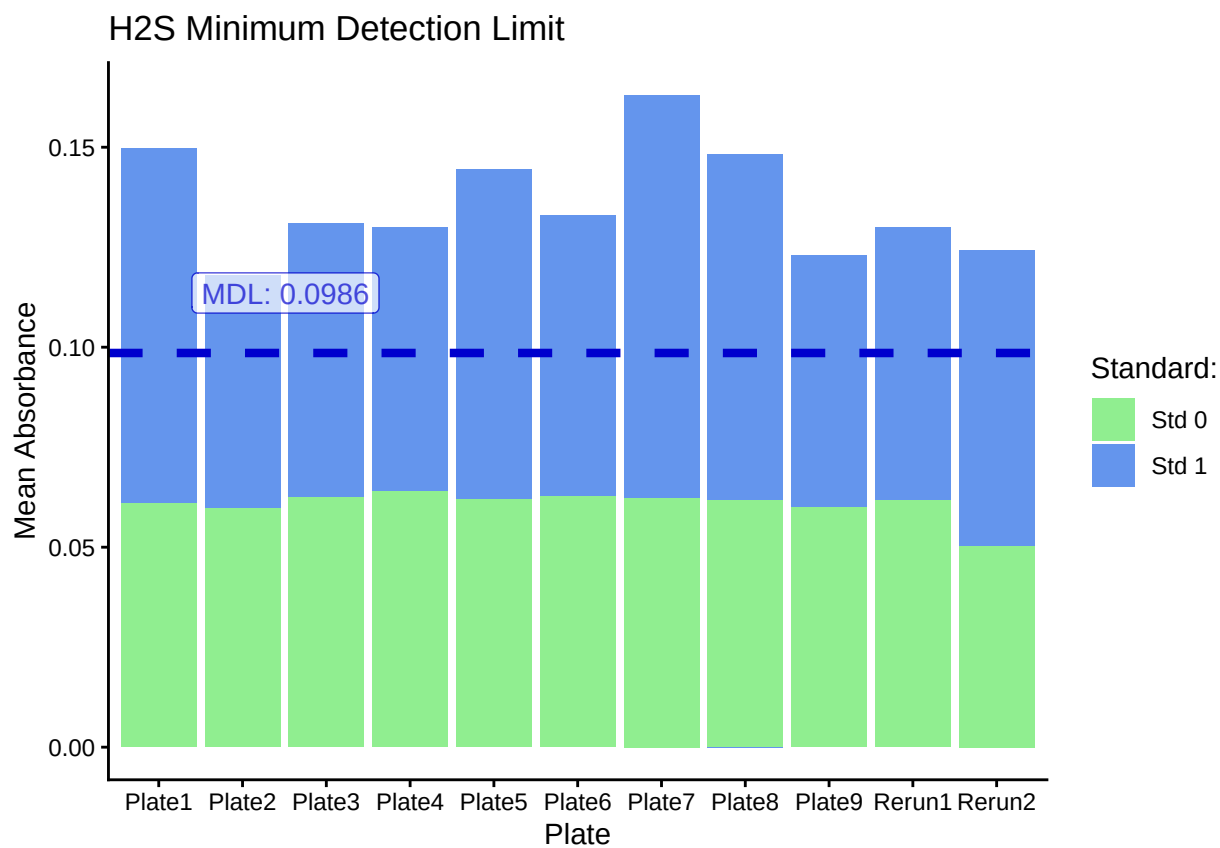
Return Flags For Standard Curve

Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate2	R2_Good	CV_Good
Plate3	R2_Bad	CV_Good
Plate3	R2_Bad	CV_High
Plate4	R2_Bad	CV_Good
Plate5	R2_Bad	CV_Good
Plate5	R2_Bad	CV_High
Plate6	R2_Good	CV_Good
Plate7	R2_Good	CV_Good
Plate8	R2_Bad	CV_Good
Plate9	R2_Good	CV_Good
Rerun1	R2_Good	CV_Good
Rerun2	R2_Good	CV_Good

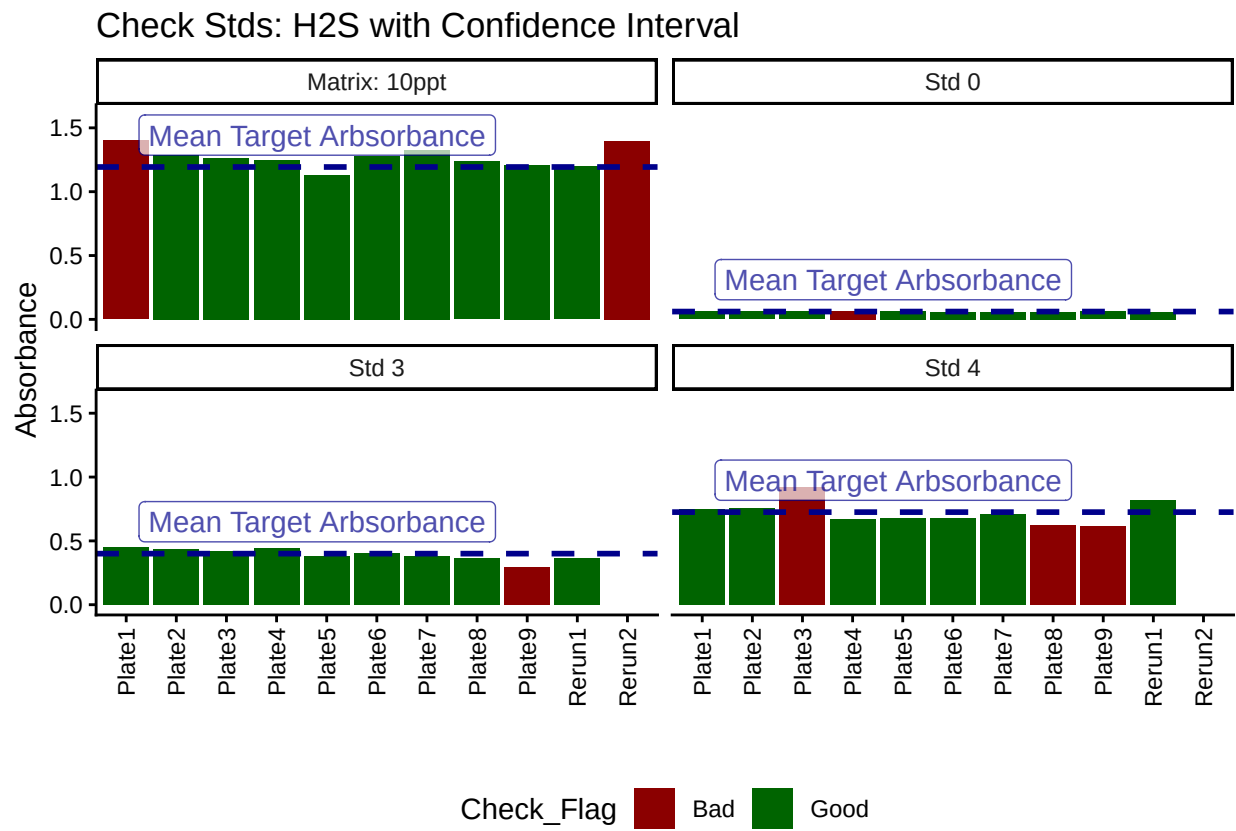
Plate	Curve_RMVD
Plate1	KEPT
Plate2	KEPT
Plate3	REMOVED
Plate4	REMOVED
Plate5	REMOVED
Plate6	KEPT
Plate7	KEPT
Plate8	REMOVED
Plate9	KEPT

Plate	Curve_RMVD
Rerun1	KEPT
Rerun2	KEPT

Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

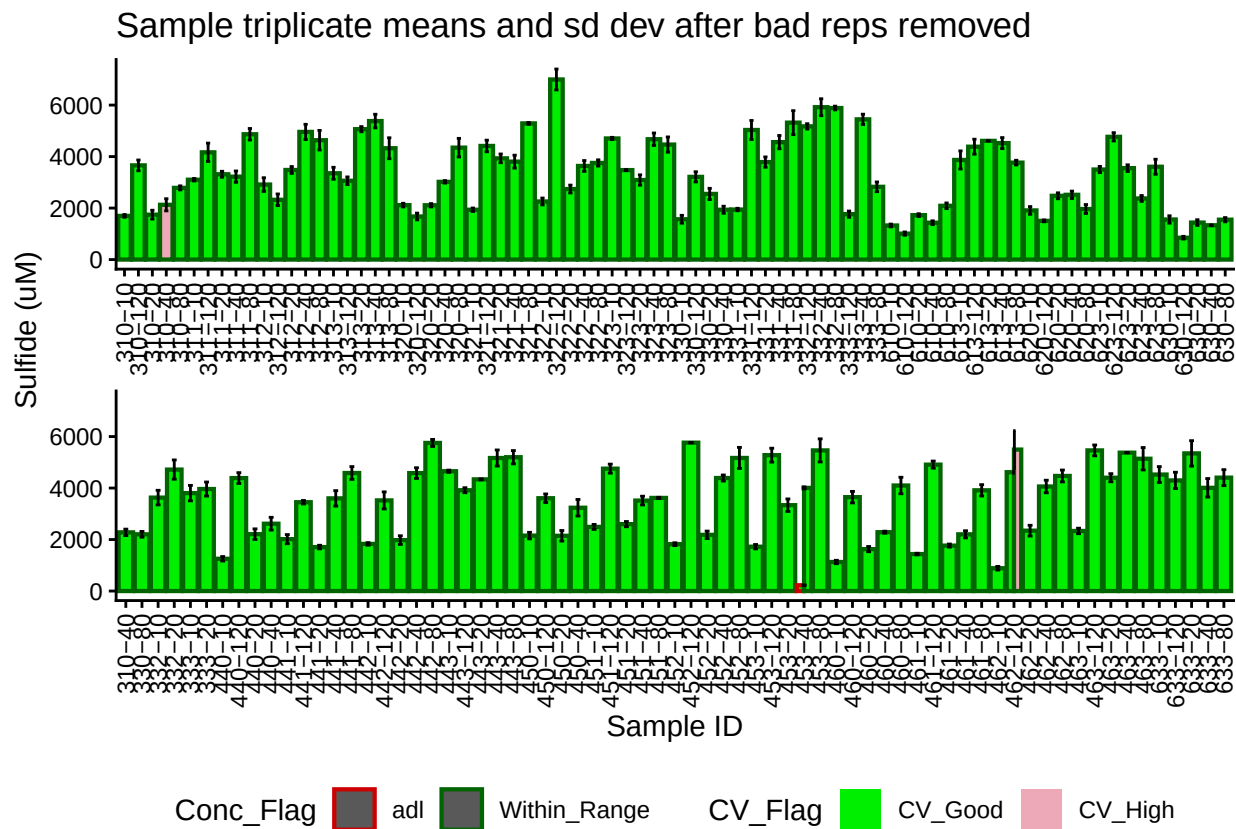
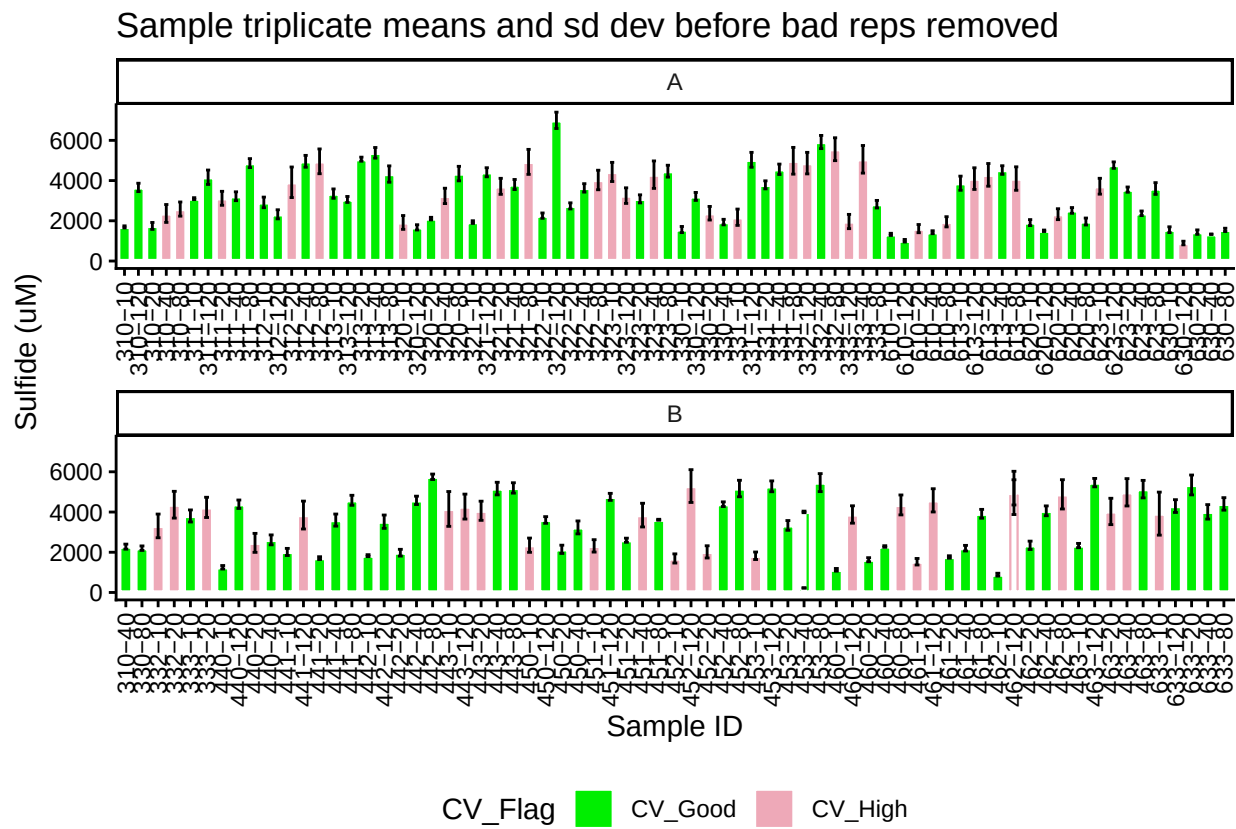
Plate	CHKs_Flag
Plate1	Std5Chk_Bad
Plate3	Std4Chk_Bad
Plate4	Std0Chk_Bad
Plate8	Std4Chk_Bad
Plate9	Std3Chk_Bad
Plate9	Std4Chk_Bad
Rerun2	Std5Chk_Bad

Plate	CHK_CV_Flags
Plate1	Std4Chk_CV_High
Plate3	Std4Chk_CV_High

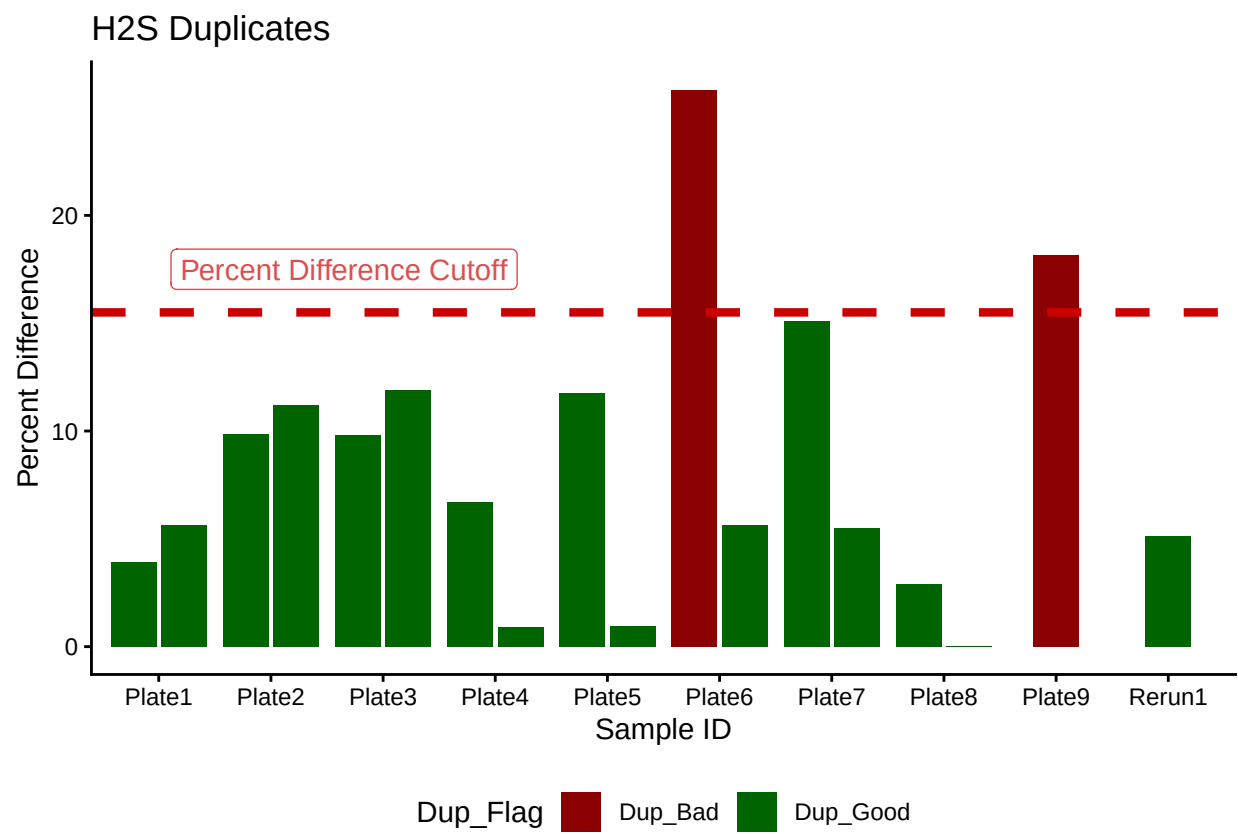
##Calculate Sulfide Concentrations & Add Concentration Flags

##Remove High CV Samples and Average

Plot Samples



Duplicate QAQC

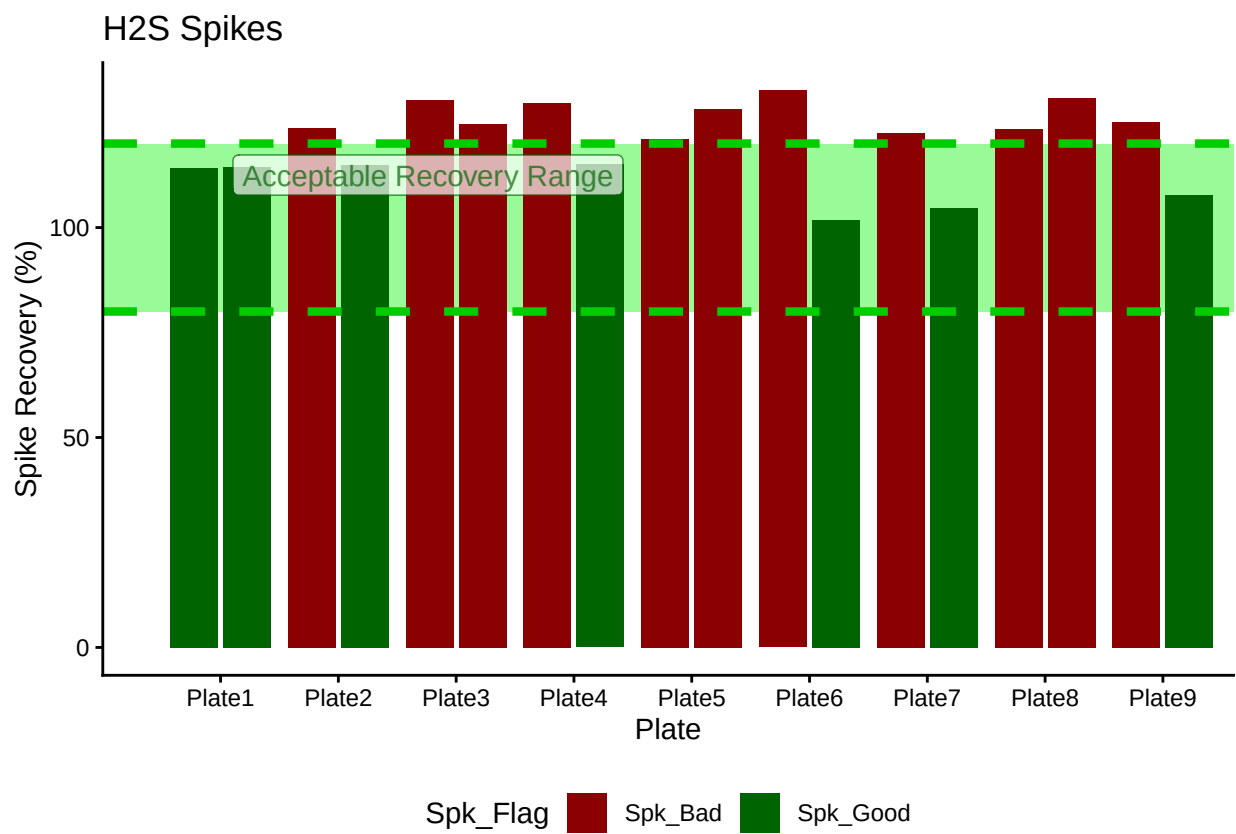


Display Dup Flags

[1] ">60% Dups Pass"

Plate	Dup_Flag
Plate1	Dup_Good
Plate2	Dup_Good
Plate3	Dup_Good
Plate4	Dup_Good
Plate5	Dup_Good
Plate6	Dup_Bad
Plate7	Dup_Good
Plate8	Dup_Good
Plate9	Dup_Bad
Rerun1	Dup_Good

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Good
Plate2	Spk_Bad
Plate3	Spk_Bad
Plate4	Spk_Bad
Plate5	Spk_Bad
Plate6	Spk_Bad
Plate7	Spk_Bad
Plate8	Spk_Bad
Plate9	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

Merge samples with Metadata & check all samples are present

```
## Some sample IDs are missing from data.
```

```
## [1] "440-80" "450-80"
```

```
##Add Flags
```

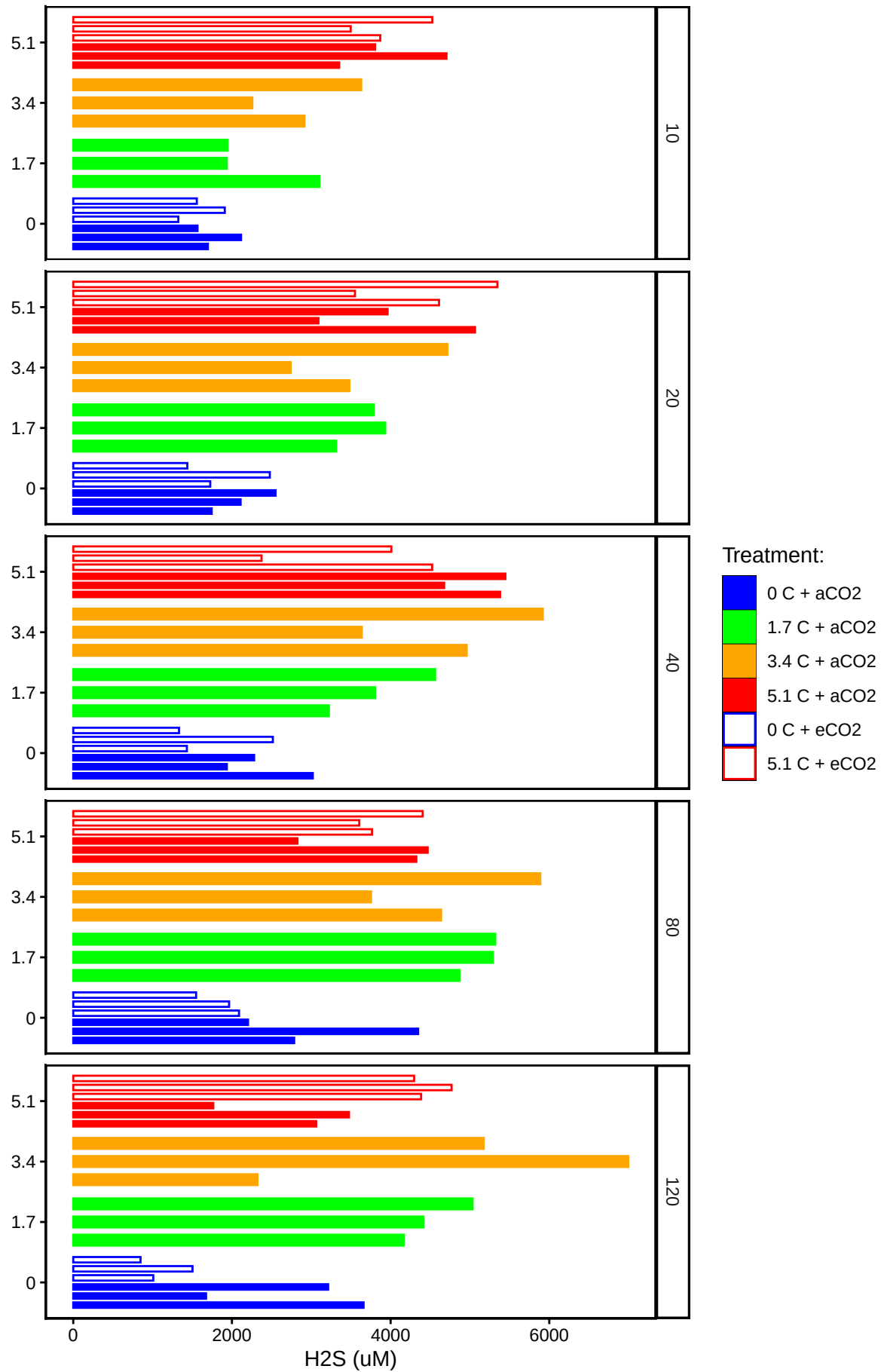
```
##Organize Data
```

```
## Adding missing grouping variables: 'Sample_ID.x'  
## Adding missing grouping variables: 'Sample_ID.x'
```

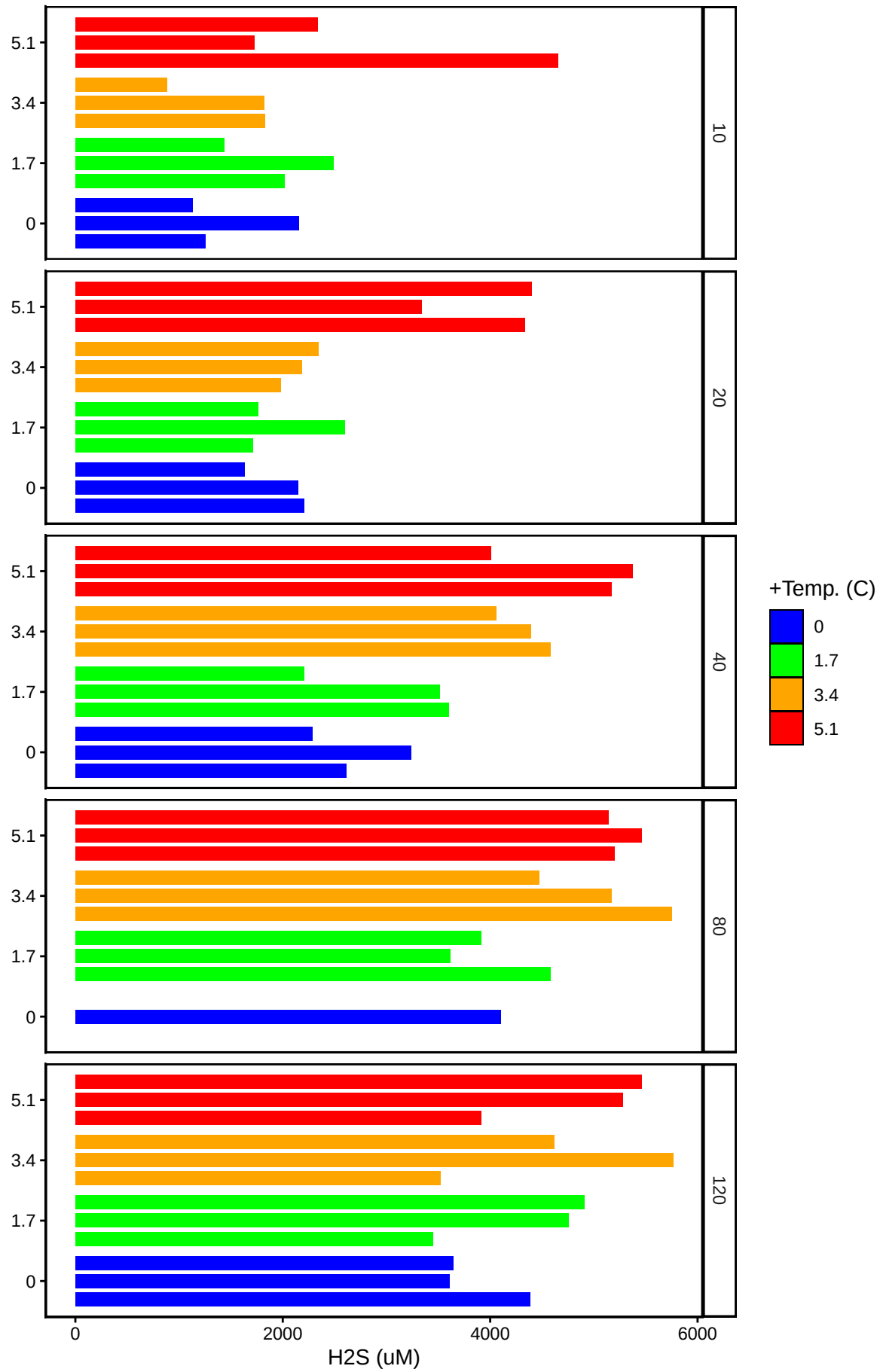
Visualize Data

```
## Visualize Data
```

C3: Porewater H2S

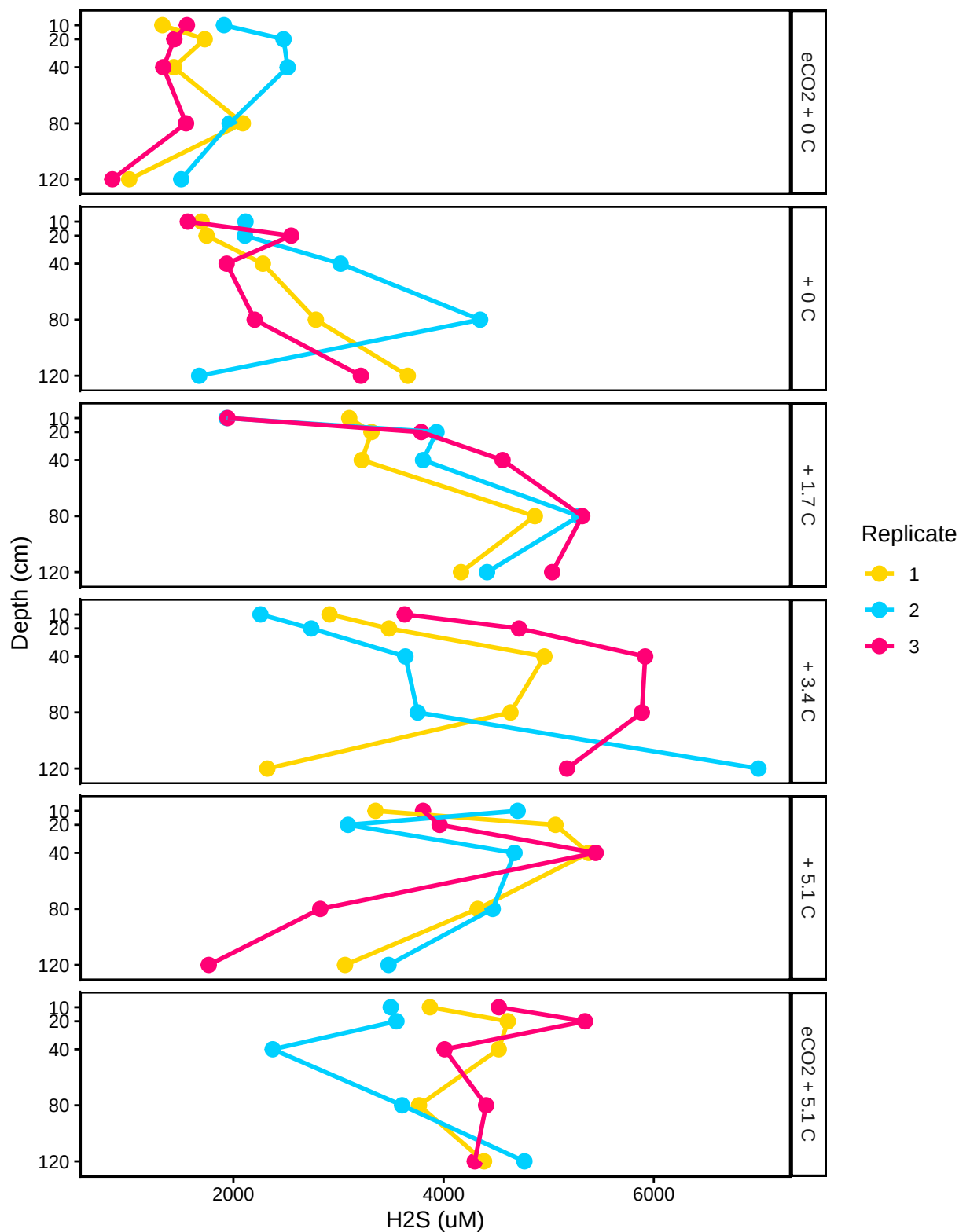


C4: Porewater H2S

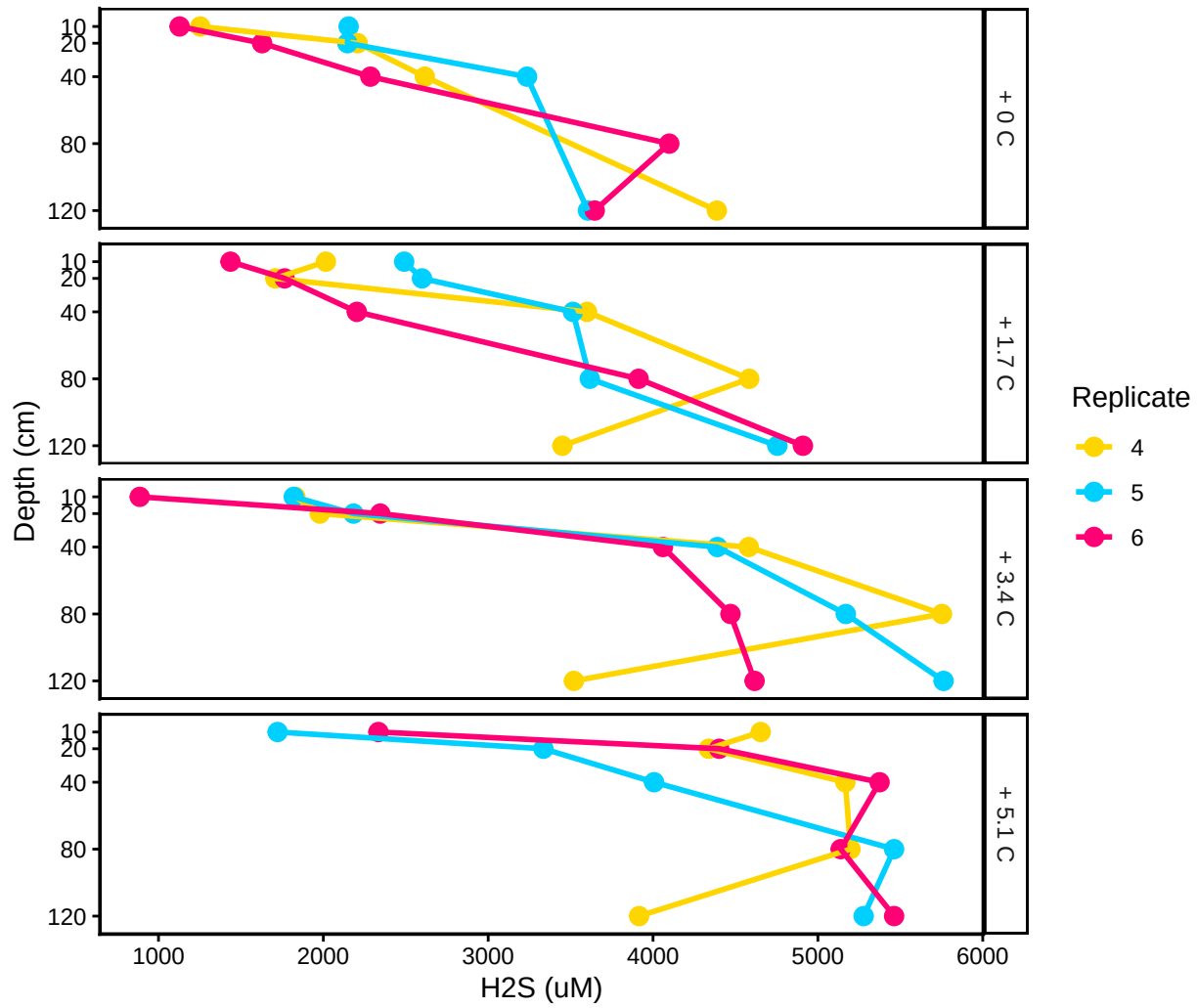


```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

C3: Porewater H₂S by Replicate



C4: Porewater H₂S by Replicate



##Export Data

END